$$Tr(\rho) = 1.$$

So far, the discussion of density matrices may have been a little abstract for some readers. Here is a worked-out example that should help bring density matrices into sharper focus. Recall the definition of Alice's density matrix from Eq. 7.20:

$$\rho_{a'a} = \sum_b \psi^*(a,b)\ \psi(a',b). \qquad (7.23)$$

Now, consider the state-vector

$$|\Psi\rangle \ = \ \frac{1}{\sqrt{2}}\Big(|ud\rangle + |du\rangle\Big).$$

Notice that two of the basis vectors have a coefficient of $\frac{1}{\sqrt{2}}$, while the other two have coefficients of zero. The state is normalized because the sum of the squared coefficients is 1. Also, all four coefficients happen to be real, which simplifies the process of complex conjugation.

Let's calculate Alice's density matrix for this state. First, for all possible inputs *a* and *b*, we'll list the values of $\psi(a, b)$. Recall that these are just the basis vector coefficients: